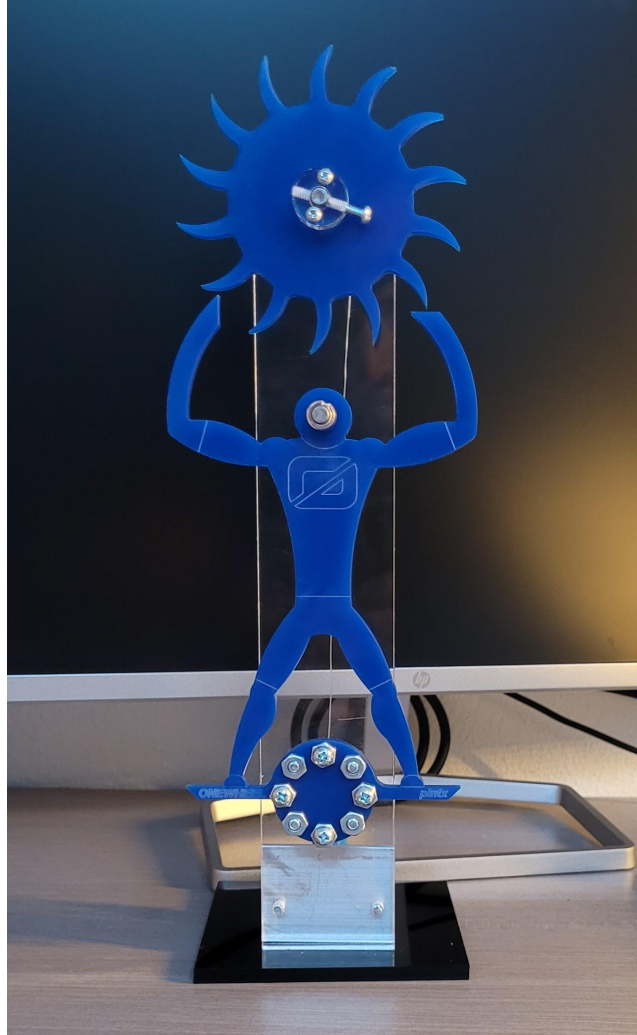




# William Harris Mechanical Clock

Section A03

## Executive summary:

The MAE 3 Mechanical Clock Project focuses on designing a mechanical clock with the objective of calculating its theoretical timing and comparing that to its experimental timing. We are given the dimensions for the escapement wheel and the pallets that interact with the escapement wheel and are tasked with designing a personalized pendulum. The pendulum I designed is in the shape of a person riding a Onewheel. I wanted it to function as close to a point mass as possible (while still being personalized), so I made my design symmetrical, with the nuts and bolts concentrated in a circle at the bottom. To calculate the theoretical timing of the clock, we use two methods; Point-Mass Analysis (PMA) and Rigid-Body Analysis (RBA). One revolution of the escapement wheel was measured to take 9.94 seconds. Out of the two, PMA came closest to this time at 8.43 seconds while the RBA resulted in 6.52 seconds. This means that PMA has an error of 15.12 percent and RBA has an error of 34.39 percent. My objective is to understand why my calculations have given me such different results than what actually occurs when running the clock.



## Theoretical Analysis:

As previously mentioned, the two methods of calculating the timing of the clock are PMA and RBA. Both have three common assumptions that friction can be neglected, that the maximum angle of motion is relatively small and that the pendulum swings freely without the influence of the escapement wheel. However, these two forms of analysis differ in that PMA assumes the entire mass of the pendulum is concentrated at one point while the RBA does not. RBA removes the assumption of mass at a single point and considers the pendulum as a rigid body with a moment of inertia along with its mass.

In physics calculations, as assumptions are removed, the analysis becomes more representative of real physics, so the RBA should turn out to be more accurate than the PMA. In fact, as a pendulum acts more and more like a point mass, both PMA and RBA should converge to the actual timing. However, this is not what my project has shown. As mentioned previously, the error of PMA was much lower than that of RBA.



## Equations & Values

$$M_{\text{total}} = M_{\text{calc}} + n * M_{\text{bolt}} [gm]$$

$$L_{\text{com}} = M_{\text{calc}} * L_{\text{acrylic}} + M_{\text{bolt}} * (L_{\text{bolt } 1} + \dots + L_{\text{bolt } n}) / M_{\text{total}} [m]$$

$$f = \omega / 2\pi \text{ hz}$$

$$T = 1/f [s]$$

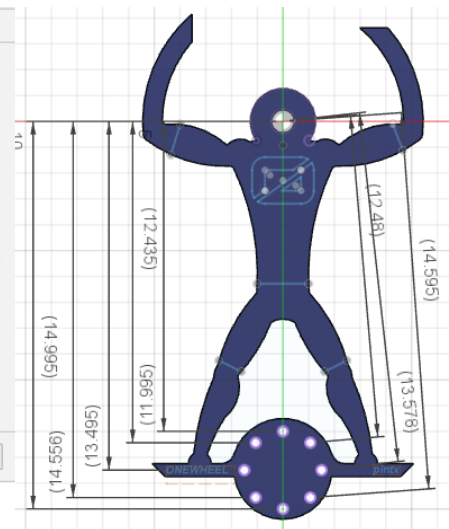
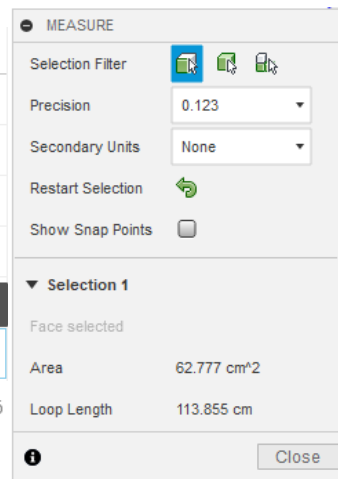
$$\omega = \sqrt{(g/L_{\text{com}})} [rad/s]$$

$$\omega = \sqrt{(M_{\text{total}} * g * L_{\text{com}} / I_{\text{total}})} [rad/s]$$

$$I_{\text{total}} = I_{\text{acrylic}} + I_{\text{bolt } 1} + \dots + I_{\text{bolt } n} [g \cdot \text{cm}^2]$$

$$\text{rps} = T * n_{\text{teeth}}$$

|                  |                                |          |
|------------------|--------------------------------|----------|
| Physical         |                                |          |
| Mass             | 47.358 g                       |          |
| Volume           | 39.864 cm^3                    |          |
| Density          | 1.188 g / cm^3                 |          |
| Area             | 197.853 cm^2                   |          |
| World X,Y,Z      | -0.001 cm, 0.318 cm, -5.991 cm |          |
| Center of Mass   | -0.001 cm, 0.318 cm, -5...     |          |
| Moment of Ine... | Iyy                            | 3428.306 |



(Total Mass)

(Length to Center of Mass)

(Natural Frequency Hz)

(Period)

(PMA Natural Frequency Rad/s)

(RBA Natural Frequency Rad/s)

(Total Moment of Inertia)

(Revolution per Second)

## Calculations:

$$M_{\text{total}} = 79.36$$

$$A_{\text{surface}} = 62.78$$

$$M_{\text{bolt}} = 4$$

$$M_{\text{acrylic}} = 47.36$$

$$L_{\text{acrylic}} = 5.99$$

$$L_{\text{com}} = 0.0902$$

$$I_{\text{acrylic}} = 3428$$

$$I_{\text{bolt}} = M_{\text{bolt}} L_{\text{bolt}}$$

$$n_{\text{bolts}} = 8$$

$$n_{\text{teeth}} = 14$$

PMA

$$\omega = 10.43 \text{ rad/s}$$

$$f = 1.66 \text{ hz}$$

$$T = 0.60 \text{ s}$$

$$\text{rps} = 8.43$$

RBA

$$I_{\text{total}} = 3861.49$$

$$\omega = 13.48 \text{ rad/s}$$

$$f = 2.15 \text{ hz}$$

$$T = 0.47 \text{ s}$$

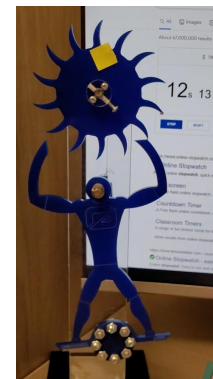
$$\text{rps} = 6.52$$

Summarizing the above equations and calculations, to find the timing of one revolution of the escapement wheel using PMA, I found the mass and center of mass properties of the pendulum and the nuts and bolts to get the total mass ( $M_{\text{total}} = 79.36 \text{ g}$ ) and center of mass ( $L_{\text{com}} = 0.0902 \text{ m}$ ). I used these to calculate the natural frequency of the pendulum in radians per second ( $\omega = 10.43 \text{ rad/s}$ ), converted that to hertz ( $f = 1.66 \text{ Hz}$ ), and took the reciprocal of the frequency in hertz to find the period of one pendulum swing ( $T = 0.60\text{s}$ ). I then multiplied the frequency by the number of teeth on the escapement wheel to get the result of **8.43 rotations per second**.

Using RBA, I found the moment of inertia properties of the pendulum and the nuts and bolts at their positions and used them to calculate the total moment of inertia ( $I_{\text{total}} = 3861.49$ ). I now used a different equation to find the frequency in radians per second ( $\omega = 13.48 \text{ rad/s}$ ), and repeated the same steps as above to get the following results; ( $f = 2.15 \text{ Hz}$ ), ( $T = 0.47\text{s}$ ), & **revolutions per second = 6.52**).

## Experimental Results:

To find the timing of the clock experimentally I took a slow-motion video of the clock running next to a monitor displaying a stopwatch so that I could note the exact time it takes the escapement wheel to make a full revolution. Of course, because of human error in starting the stopwatch at the same time as the clock and the video, I had to subtract the time the revolution started from the time that it ended to get the result of 9.94 seconds per revolution.



## Discussion:

Despite the fact that RBA is supposed to be more accurate, it turned out to have a greater error than PMA. I am not entirely sure why this has occurred because even if my pendulum acts more like a point mass, it does not change that RBA removes an assumption that PMA does not, therefore emulating real physics more accurately.

My current thoughts are that if I measured the period of the pendulum without the escapement wheel interaction, my measurements would be closer to that of the theoretical results due to the lack of friction caused by the interaction. However, I do think it is possible I may have just made an error in my calculations. I checked multiple times and compared it to other classmates, and have found no errors, but this is still a possibility.

The fact that RBA turned out the lowest value, followed by PMA, and then the real-life results tell me that RBA might be telling me what the timing would be in perfect conditions and PMA may be closer to the experimental value due to friction in the system.

## Appendix:

| Variable Description                                                      | Variable Name    | Values/Equations                                                                                       | Units             |
|---------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------|-------------------|
| <b>Acrylic Pendulum Specifications</b>                                    |                  |                                                                                                        |                   |
| Area                                                                      | A                | 62.777                                                                                                 | cm <sup>2</sup>   |
| Thickness                                                                 | t                | 0.635                                                                                                  | cm                |
| Volume                                                                    | Vol              | $A * t = 39.863395$                                                                                    | cm <sup>3</sup>   |
| Density                                                                   | p                | $Vol * p = 1.188$                                                                                      | g/cm <sup>3</sup> |
| Calculated Mass of Acrylic                                                | M_Calc           | 47.35771326                                                                                            | g                 |
| Actual Mass of Acrylic                                                    | M_Act            | 47.358                                                                                                 | g                 |
| Length to Center of Mass of Acrylic                                       | La               | 5.991                                                                                                  | cm                |
| Percent Error in Acrylic Mass Calculation                                 |                  | $(M\_Act - M\_Calc) / M\_Act * 100 = 0.0006054732041$                                                  | %                 |
| <b>Calculate Total Mass of Pendulum</b>                                   |                  |                                                                                                        |                   |
| Mass of One Bolt with Two Nuts                                            | Mb               | 4                                                                                                      | g                 |
| Number of Bolts with Two Nuts                                             | Nb               | 8                                                                                                      |                   |
| Total Mass of Pendulum with Nuts and Bolts                                | Mt               | $M\_Calc + Mb * Nb = 79.35771326$                                                                      | g                 |
| <b>Calculate Center of Mass of Pendulum with Bolts</b>                    |                  |                                                                                                        |                   |
| Length to Center of Mass of Bolt 1                                        | L_b1             | 11.995                                                                                                 | cm                |
| Length to Center of Mass of Bolt 2                                        | L_b2             | 12.435                                                                                                 | cm                |
| Length to Center of Mass of Bolt 3                                        | L_b3             | 12.435                                                                                                 | cm                |
| Length to Center of Mass of Bolt 4                                        | L_b4             | 13.495                                                                                                 | cm                |
| Length to Center of Mass of Bolt 5                                        | L_b5             | 13.495                                                                                                 | cm                |
| Length to Center of Mass of Bolt 6                                        | L_b6             | 14.556                                                                                                 | cm                |
| Length to Center of Mass of Bolt 7                                        | L_b7             | 14.556                                                                                                 | cm                |
| Length to Center of Mass of Bolt 8                                        | L_b8             | 14.995                                                                                                 | cm                |
| Length to Center of Mass in Meters                                        | Lcom_meter       | $(M\_Calc * La + C17 * (C22 + C23 + C24 + C25 + C26 + C27 + C28 + C29)) / (C19 * 100) = 0.09016994451$ | m                 |
| Estimated Center of Mass of Pendulum with Nuts and Bolts                  | Lcom_est         | 0.095                                                                                                  | m                 |
| Percent Error in Pendulum Nuts and Bolts Lcom Estimate                    | Lcom_error       | $abs(Lcom\_meter - Lcom\_est) * 100 = 0.4830055486$                                                    | m                 |
| <b>Calculate Natural Frequency and Timing using Point Mass Assumption</b> |                  |                                                                                                        |                   |
| Gravitational Constant                                                    | grav             | 9.81                                                                                                   | m/s <sup>2</sup>  |
| Natural Frequency in radians/sec                                          | nat_freq_rad_sec | $sqrt(grav / Lcom\_meter) = 10.43046338$                                                               | rad/s             |
| Natural Frequency in Hz                                                   | nat_freq_hz      | $nat\_freq\_rad\_sec / (2 * PI()) = 1.660059805$                                                       | hz                |
| Period of Oscillation                                                     | period           | 0.6023879363                                                                                           | s                 |
| Number of Teeth on Escapement Wheel                                       | nteeth           | 14                                                                                                     |                   |
| Calculated Time of One Revolution of                                      | time_calc        | $period * nteeth = 8.433431108$                                                                        | s                 |

|                                                                           |                     |                                                                                                       |                   |
|---------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------|-------------------|
| Escapement Wheel                                                          |                     |                                                                                                       |                   |
| Measured Time of One Revolution of Escapement Wheel                       | time_meas           | 9.9439 s                                                                                              |                   |
| Percent Error in Clock Timing                                             | time_error          | $(\text{time\_meas} - \text{time\_calc}) / \text{time\_meas} * 100 = 15.18990428$                     | %                 |
| <b>Calculate Natural Frequency and Timing using Rigid Body Assumption</b> |                     |                                                                                                       |                   |
| Moment of Inertia of Pendulum                                             | I_a                 | 3428.306                                                                                              | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 1                                               | I_b1                | $M_b * 11.995 = 47.98$                                                                                | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 2                                               | I_b2                | $M_b * 12.48 = 49.92$                                                                                 | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 3                                               | I_b3                | $M_b * 12.48 = 49.92$                                                                                 | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 4                                               | I_b4                | $M_b * 13.578 = 54.312$                                                                               | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 5                                               | I_b5                | $M_b * 13.578 = 54.312$                                                                               | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 6                                               | I_b6                | $M_b * 14.595 = 58.38$                                                                                | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 7                                               | I_b7                | $M_b * 14.595 = 58.38$                                                                                | g·cm <sup>2</sup> |
| Moment of Inertia of Bolt 8                                               | I_b8                | $M_b * 14.995 = 59.98$                                                                                | g·cm <sup>2</sup> |
| Total Moment of Inertia                                                   | I_total             | $\text{sum}(C45:C53) = 3861.49$                                                                       | g·cm <sup>2</sup> |
| Natural Frequency in radians/sec                                          | rb_nat_freq_rad_sec | $= \text{sqrt}(M_t * \text{grav} * L_{\text{com\_meter}} * 10000 / (I_{\text{total}})) = 13.48287537$ | rad/s             |
| Natural Frequency in Hz                                                   | rb_nat_freq_hz      | $= \text{rb\_nat\_freq\_rad\_sec} / (2 * (\pi)) = 2.145866262$                                        | hz                |
| Period of Oscillation                                                     | rb_period           | $1 / \text{rb\_nat\_freq\_hz} = 0.4660122663$                                                         | s                 |
| Calculated Time of One Revolution of Escapement Wheel                     | rb_time_calc        | $\text{rb\_period} * \text{n\_teeth} = 6.524171728$                                                   | s                 |
| Percent Error in Clock Timing                                             | rb_time_error       | $(\text{time\_meas} - \text{rb\_time\_calc}) / \text{time\_meas} * 100 = 34.39021181$                 | %                 |